

Kartik Gupta

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📁 Portfolio

Computational scientist with 6+ years of HPC-driven CFD experience, specializing in parallelized solver development (MPI/OpenMP), fluid-structure interaction, and large-scale performance optimization — complemented by hands-on ML/AI expertise in LLM deployment, RAG pipelines, and deep learning on HPC clusters.

EDUCATION AND TRAINING

Ph.D. (Engineering and Applied Science, University of Massachusetts Dartmouth (U.S.A.))	2023 – Present
M.Tech. (Data Science, University of Massachusetts Dartmouth (U.S.A.))	2025 – 2026
M.Tech. (Mechanical and Process Engineering, TU Bergakademie Freiberg (Germany))	2019 – 2022
B.Tech. (Mechanical Engineering, S.R.M. Institute of Science and Technology (India))	2014 – 2018

TECHNICAL EXPERIENCE

Teaching Assistant (University of Massachusetts Dartmouth)

High Performance Scientific Computing

January 2026 – May 2026

- Guided students in Python, C, numerical linear algebra, and parallel computing (MPI, OpenMP, CUDA), while teaching Git (version control), Bitbucket, and Jupyter-based data analysis.
- Mentored Data Science students by holding weekly computational help-desk hours to develop and optimize parallel programs for achieving significant speedups (using GPUs), covering memory access, load balancing, debugging (GDB, TotalView), and performance/scalability analysis.
- Assisted in debugging and profiling parallel codes using GDB, TotalView, and performance analysis tools.
- Delivered technical presentations and clearly articulated complex computational concepts to diverse student audiences.

Teaching Assistant (University of Massachusetts Dartmouth)

Thermal Systems Design (MNE-421)

September 2025 – December 2025

- Assisted engineering students in thermal system design and analysis using EES and COMSOL Multiphysics.
- Assisted teams in planning workflows, meeting deadlines, and resolving technical challenges.
- Provided feedback on calculations and design reports, ensuring adherence to performance, cost, and feasibility constraints.

Graduate Researcher (Data Scientist) (University of Massachusetts Dartmouth)

Machine Learning / AI

January 2025 – Present

- Deployed Ollama LLM on a Unity HPC cluster via Docker with GPU acceleration, achieving $3\text{--}5\times$ inference throughput improvement over CPU baselines; profiled end-to-end latency and memory footprint to identify optimization opportunities analogous to TensorRT-LLM deployment workflows.
- Built and benchmarked a production-ready RAG-based LLM chatbot using HuggingFace embeddings, FAISS vector search, and REST APIs on AWS; improved response relevance by 40% through prompt engineering, retrieval scheduling, and inference pipeline optimization — directly applicable to GenAI inference serving (vLLM/SGLang patterns).
- Developed CNN-based image classification pipeline (Xception / ImageNet transfer learning, TensorFlow/Keras) for brain tumor detection; applied INT8-equivalent quantization-aware training strategies, data augmentation, and cross-validation to optimize model size and throughput for clinical AI pipelines.
- Built time-series regression models (ARIMA, XGBoost) to forecast COVID-19 trends from epidemiological data; reduced MSE by 30% via feature engineering and hyperparameter tuning on temporal waveform data.

Research Assistant (University of Massachusetts Dartmouth)

High-Performance Multiphase CFD Simulations

January 2023 – September 2025

- Developed a C++ based fully Eulerian multiphase CFD solver for water-entry problems, incorporating Volume of Fluid (VOF) and fictitious domain methods for fluid–structure interaction (FSI).
- Designed and parallelized the solver using MPI on Linux HPC clusters; implemented Bi-Conjugate Gradient Stabilized (BiCGStab) and multigrid methods for efficient solution of the pressure Poisson equation, achieving an $8\times$ improvement in accuracy with residuals reduced to 10^{-12} .

- Optimized multi-core scalability through domain decomposition, load balancing, and communication-efficient strategies over InfiniBand interconnects; conducted strong and weak scaling analyses, resulting in a $6\times$ performance improvement.
- Conducted performance benchmarking and scalability analysis across core counts, comparable to roofline analysis used in GPU inference optimization.
- Investigated effects of nose curvature on vertical water entry, demonstrating how concave, flat, and convex noses influence cavity formation, splash curtain dynamics, and impact forces.
- Showed that oblique water entry produces asymmetric cavity shapes, reduced impact forces, and lower added mass; developed methodology to quantify added mass for different nose geometries and a theoretical model to predict cavity closure (pinch-off) events.

Research Assistant (TU Bergakademie Freiberg, Germany)

High-Performance Multiphase CFD Simulations

May 2021 – August 2022

- Performed transient multiphase simulations in OpenFOAM using VOF, dynamic mesh, and Adaptive Mesh Refinement (AMR); investigated aerosol atomization in vocal folds using dynamic (moving) mesh techniques and Bash scripting to model vocal cord oscillations.
- Optimized solver for high density-ratio flows, gaining deep intuition for numerical stability and performance trade-offs applicable to low-latency, resource-constrained inference systems.
- Conducted bionic modelling of Archer fish, investigating water jet behavior related to the Plateau–Rayleigh instability using OpenFOAM and ANSYS Fluent.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, MATLAB, L^AT_EX, Bash, Linux.
- **HPC / Parallel Computing:** Slurm, MPI, OpenMP, CUDA; strong and weak scaling analysis, domain decomposition, load balancing, InfiniBand networking.
- **Machine Learning / AI:** TensorFlow, Keras, PyTorch, JAX, HuggingFace Transformers, LLM fine-tuning, RAG pipelines, FAISS, AWS, OpenAI API; pandas, NumPy, SciPy, Matplotlib, XGBoost.
- **CFD / Simulation Software:** OpenFOAM, ANSYS Fluent, StarCCM+, COMSOL Multiphysics, ParaView (pvpypython), CGNS/HDF5 data pipelines.
- **CAD & Visualization:** SolidWorks, AutoCAD, CATIA, Blender, OpenGL.
- **Debugging & Profiling:** GDB, TotalView, performance/scalability benchmarking.
- **Dev Tools:** Git, Bitbucket, Docker, VSCode, Jupyter Notebooks.

PROJECTS

1. Image Classification with CNN on CIFAR-10

 GitHub

Engineered a production-grade computer vision pipeline by fine-tuning a pretrained **ResNet18** backbone (ImageNet weights) on CIFAR-10 via **transfer learning** in PyTorch, replacing the classification head for 10-way inference. Implemented a full MLOps-style training loop featuring test-time augmentation (random resized crop, horizontal flip, color jitter, rotation), **StepLR** learning-rate annealing, validation-driven best-checkpoint serialization, and a **GitHub Actions** CI/CD pipeline for automated regression testing. Achieved **~92.8% top-1 test accuracy**, demonstrating strong domain adaptation from large-scale pretraining to low-resolution benchmark data.

2. Fluid Dynamics Solver — 2D Incompressible Navier–Stokes (MPI-Parallel)

 GitHub

Developed a research-grade 2D incompressible CFD solver from scratch in Python/NumPy, implementing a **fractional-step pressure-projection** time-marching scheme for the Navier–Stokes equations on a structured Cartesian grid with ghost-cell boundary conditions. Parallelized the iterative pressure Poisson solve using **MPI collectives** (mpi4py all-reduce) for distributed-memory scalability; wrote simulation output to **HDF5/CGNS** format via h5py and automated post-processing visualizations (streamplot, quiver) in Matplotlib. Validated solver correctness against lid-driven cavity benchmarks with a comprehensive **pytest** suite and CI integration.

3. Credit Card Fraud Detection — Imbalanced Binary Classification

 GitHub

Designed and trained a **deep MLP** fraud detection model in PyTorch, addressing severe class imbalance ($\ll 1\%$ fraud prevalence) through **weighted loss** (BCEWithLogitsLoss with computed **pos_weight**), replicating industry-standard techniques used in fintech anomaly detection pipelines. Leveraged **torch.compile** (Inductor backend) for kernel fusion and GPU acceleration, with an 80/10/10 stratified train/val/test split, Adam optimizer, and patience-based **early stopping** to prevent overfitting on minority class. Delivered a fully reproducible, CI-gated codebase with a **pytest** unit-test suite covering model architecture, data pipeline integrity, and training loop contracts.

4. **DNN Surrogate Model for Hydrodynamic Force Prediction**  GitHub
Built and trained a **physics-informed surrogate model** — a 7-layer deep MLP (3→128→256→512→512→256→128→3) in PyTorch — on 1M+ synthetic CFD samples to predict drag, lift, and torque on parameterized ellipsoidal hull geometries across a wide Reynolds-number design space, reducing inference latency from **hours-to-days** (full Navier–Stokes solve) to **milliseconds**. Applied z-score feature normalization, **outlier filtering**, Adam optimization (lr = 1e-5), and patience-based early stopping; accelerated training via CUDA and `torch.compile` (Inductor). Shipped a pre-trained **checkpoint** (`best_model.pth`) enabling zero-retraining deployment for rapid parametric design sweeps in naval architecture and marine engineering workflows.
5. **Added Mass Force Prediction — Dual PyTorch & JAX ML Surrogate**  GitHub
Developed a **data-driven regression surrogate** for real-time added mass force prediction in fluid–structure interaction (FSI) transient events (water entry, wave slamming), eliminating the need for computationally prohibitive Navier–Stokes solves during iterative structural design. Delivered two fully independent, framework-benchmarked implementations: **PyTorch** (`torch.compile`, CUDA, multi-worker DataLoaders) and **JAX** (JIT-compiled mini-batch update via Optax), sharing identical MLP architecture (3→128→128→64→1), Kaiming weight initialization, MSE loss, and early stopping — confirming framework-agnostic reproducibility. Results benchmarked with full training curves in companion PDF reports; pytest suites cover forward-pass correctness, normalization, data masking, and DataLoader integrity.
6. **Multi-Agent LLM Framework for Code Translation & Automated Debugging**  GitHub
 *PyPI release upcoming*
Architected a **production-grade agentic code migration system** using **LangGraph** state graphs, decomposing cross-language translation into five orchestrated agents: semantic understanding, transpilation, gcc-based compilation & execution, error-driven debugging, and a logical correctness reviewer — with conditional feedback edges structurally analogous to **iterative residual correction** in numerical solvers. Benchmarked on six representative test cases (recursion, dynamic memory allocation, string manipulation, tuple unpacking, pointer arithmetic, floating-point integration), achieving **100% compilation success** and **97.5% logical-correctness rate** vs. 58.3%/49.2% for single-pass GPT-4 baseline; average convergence in **1.38 iterations** with a bounded 2K+3 LLM-call budget per task. Stack: **LangGraph**, Python subprocess; PyPI packaging in progress.
7. **Agentic RAG — Stateful, Decision-Driven Retrieval-Augmented Generation**  GitHub
Engineered a **production-grade agentic RAG system** using **LangGraph** finite-state orchestration, replacing brittle single-pass retrieval chains with a **multi-checkpoint inference pipeline**: document ingestion (PDF/TXT/MD) into a **FAISS** vector store (OpenAI `text-embedding-3-small`), top-*k* semantic retrieval, LLM-based *relevance grading*, *hallucination/grounding verification*, conditional retry routing, safe fallback, and a **SymPy** symbolic equation-solving tool for scientific Q&A. Achieves superior answer fidelity and **reduced hallucination rate** vs. naive RAG by enforcing explicit quality gates before trusting generated output; deployed end-to-end with a **Streamlit** chat interface. Fully test-covered with a pytest suite validating graph routing, grounding logic, tool dispatch, and FAISS ingestion correctness.
8. **Brain Tumor MRI Classification — Clinical-Grade Deep Learning Pipeline**  GitHub
Developed an end-to-end **medical image analysis** pipeline for multi-class brain tumor classification from MRI scans, fine-tuning a pretrained **ResNet18** backbone (ImageNet weights) with a task-specific classification head via **transfer learning** in PyTorch. Engineered a robust data ingestion layer (**TumorDataset**) with **augmentation regularization** (random resized crop, horizontal flip, rotation, color jitter, ImageNet normalization), 80/20 stratified train/val split, Adam optimizer with **StepLR** decay, and validation-accuracy-driven checkpoint selection to maximize generalization on imbalanced medical datasets. Delivered production-quality model weights (`resnet18.best_model.pth`), automated **CI/CD** via GitHub Actions, and a comprehensive pytest suite validating dataset construction, label mapping, model output dimensionality, and evaluation metric contracts.

PUBLICATIONS

- L. Fritzsche, M. Heinrich, **K. Gupta**, R. Schwarze, and K. Bauer, Experimental and numerical investigation of the aerosol generation mechanisms at the human vocal folds, ILASS Europe 2023, 32nd Conference on Liquid Atomization and Spray Systems, Naples, Italy (2023).
- **K. Gupta**, and N. K. Verma, Simulation and experimental approaches for design and development of high-speed elevator (2021).
- **K. Gupta**, Mechanical behavior of aluminium (AA6061) reinforced with alumina and rice husk ash (2021).

CONFERENCE CONTRIBUTIONS

- **K. Gupta**, M. Raessi, and J. L. Belden, Computational study of water entry of spinning solid cylinders of various nose curvatures, 78th Annual Meeting of the APS's Division of Fluid Dynamics (APS/DFD), Houston, Texas (2025).

- **K. Gupta**, M. Raessi, and J. L. Belden, An approach to quantifying added mass from computational simulations of water entry of solid objects, Summer Meeting of the American Society of Mechanical Engineers (ASME/FEDSM), Philadelphia, Pennsylvania (2025).
- **K. Gupta**, M. Raessi, and J. L. Belden, Computational study of asymmetric water entry of cylindrical solid objects: Effects of tilt angle and nose curvature, 77th Annual Meeting of the APS's Division of Fluid Dynamics (APS/DFD), Salt Lake City, Utah (2024).
- **K. Gupta**, M. Raessi, and J. L. Belden, Computational simulation of water entry of solids — Effect of nose curvature on splashing dynamics and cavity closure, 76th Annual Meeting of the APS's Division of Fluid Dynamics (APS/DFD), Washington, DC (2023).

HONOURS AND AWARDS

- **First Prize Winner**, Poster Presentation Competition, MUST event, University of Massachusetts Dartmouth (2025).
- **APS Travel Grant**, American Physical Society, Division of Fluid Dynamics (2024).
- **AUMUND Scholarship**, AUMUND Fördertechnik GmbH (2021).
- **Deutschland Stipendium**, TU Bergakademie Freiberg (2020).